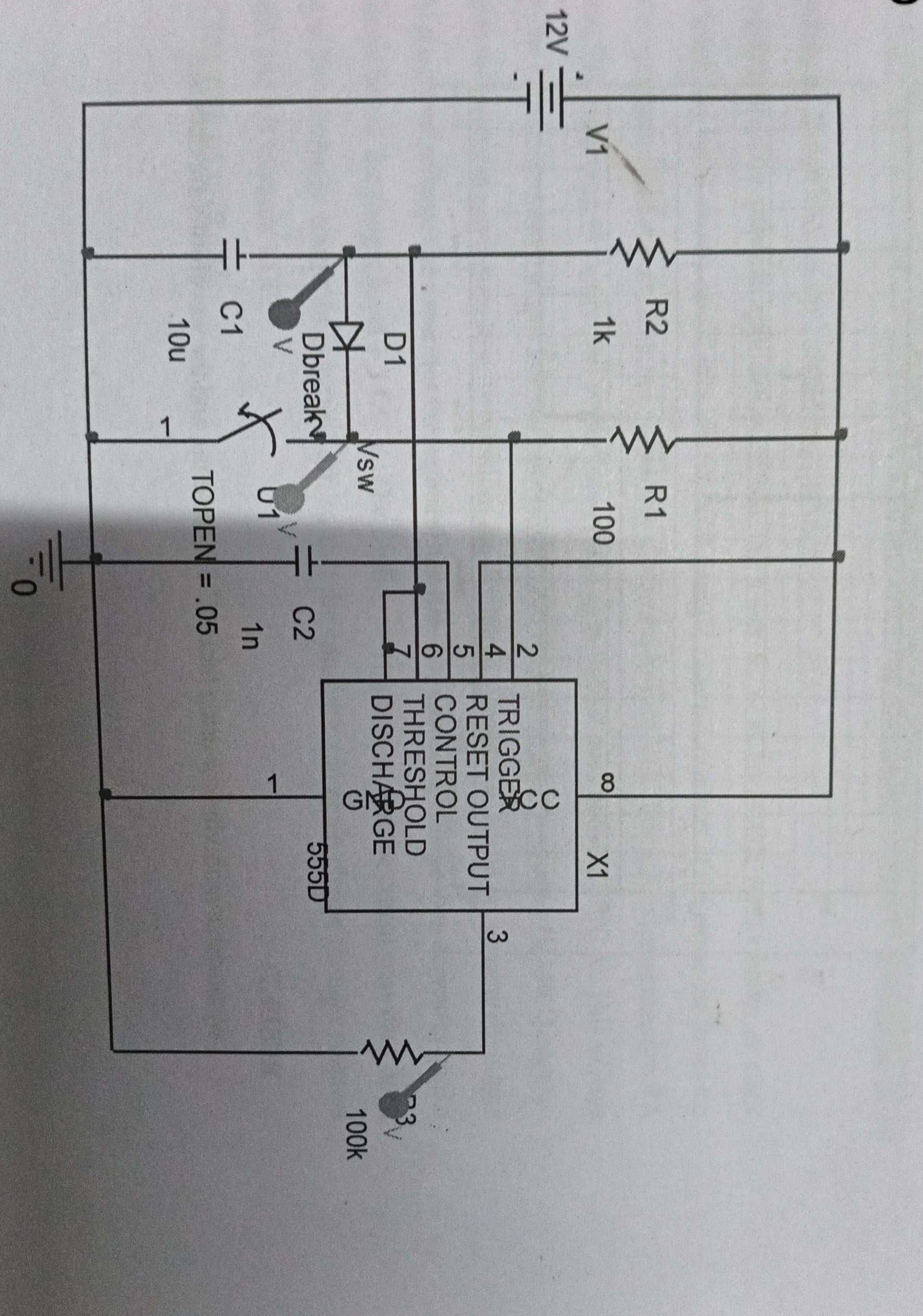
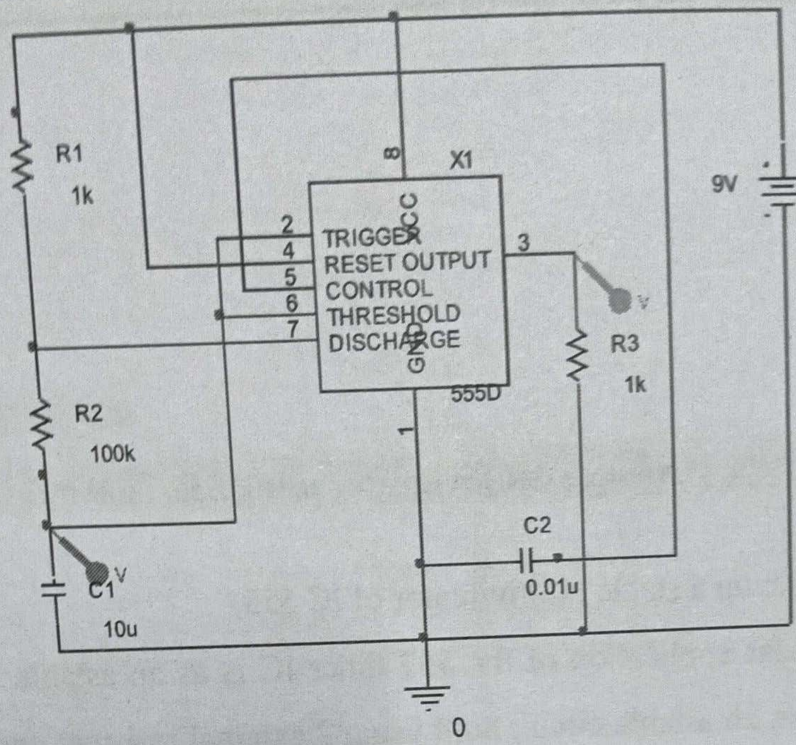
monistable multivibrator using 55

CIRCUIT DIAGRAM:



astable multivibrator using 555 tim



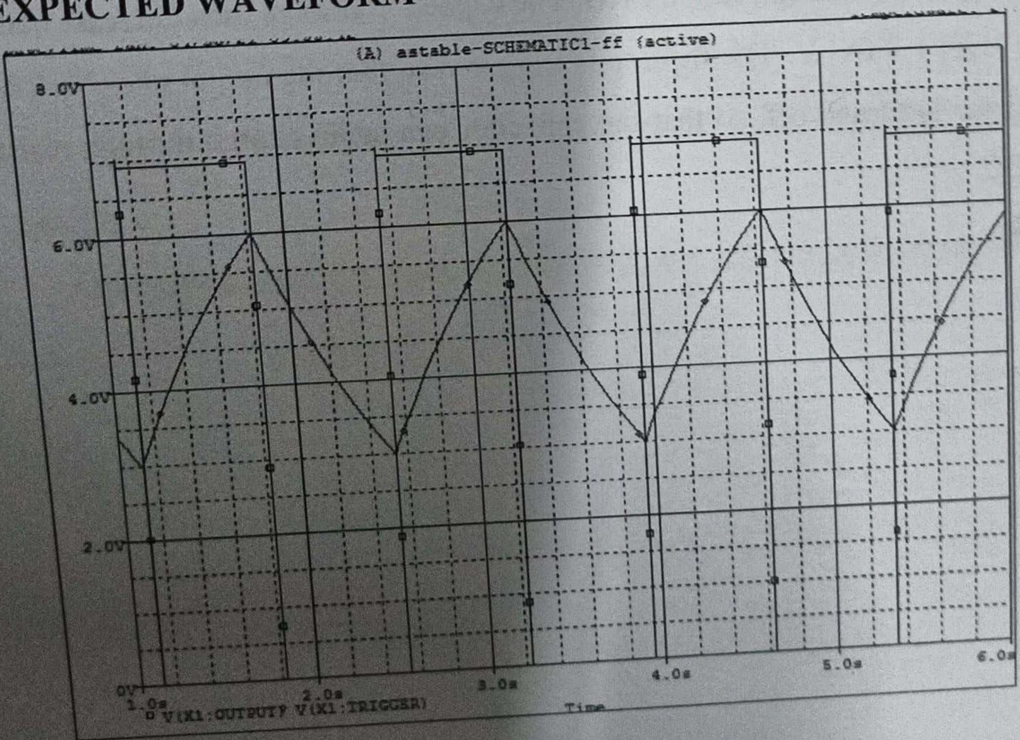
SIMULATION PROFILE:

Time domain analysis

Run to time=6s

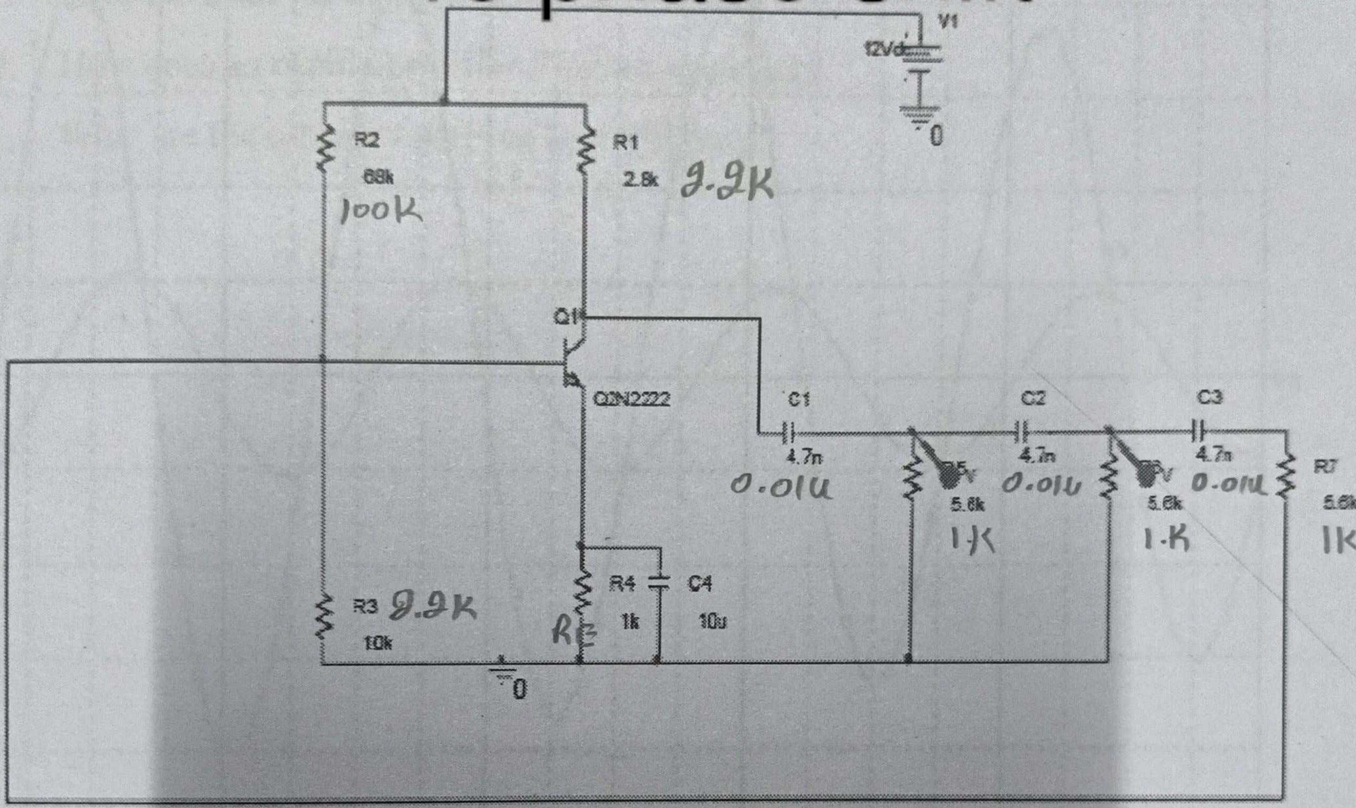
Start saving data after=1s

EXPECTED WAVEFORM



RESULT:

rc phase shift



Simulation Profile:

Time domain analysis

Run to time=1ms

Start saving data after=0ms

Maximum step size = 0

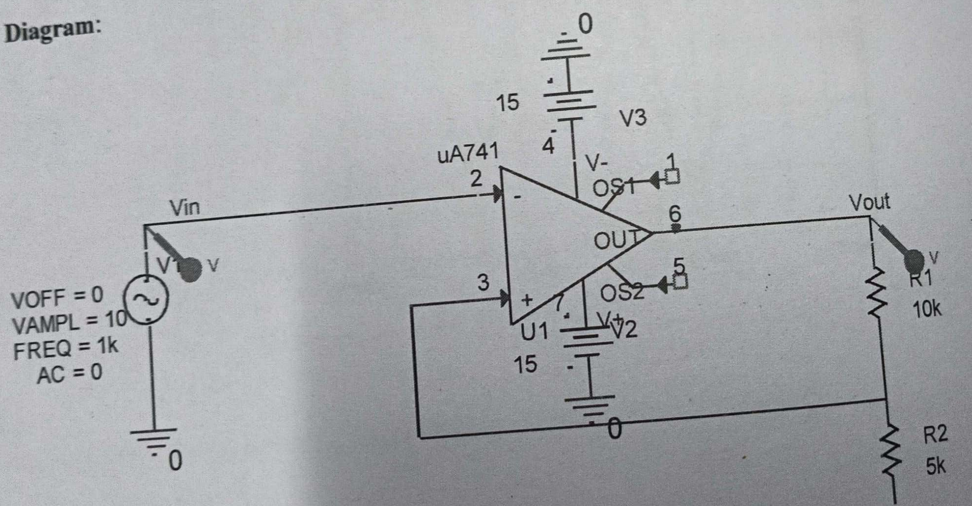
Output Waveform:

Aim: Test a comparator circuit and design a Schmitt trigger for the given UTP and LTP values and obtain the hysteresis

Theory:

A Schmitt trigger or a squaring circuit converts an irregular shaped waveform to a square wave or pulse. This also gives an output waveform virtually discontinues at the comparison voltage and it also exhibits a phenomenon called hysteresis.

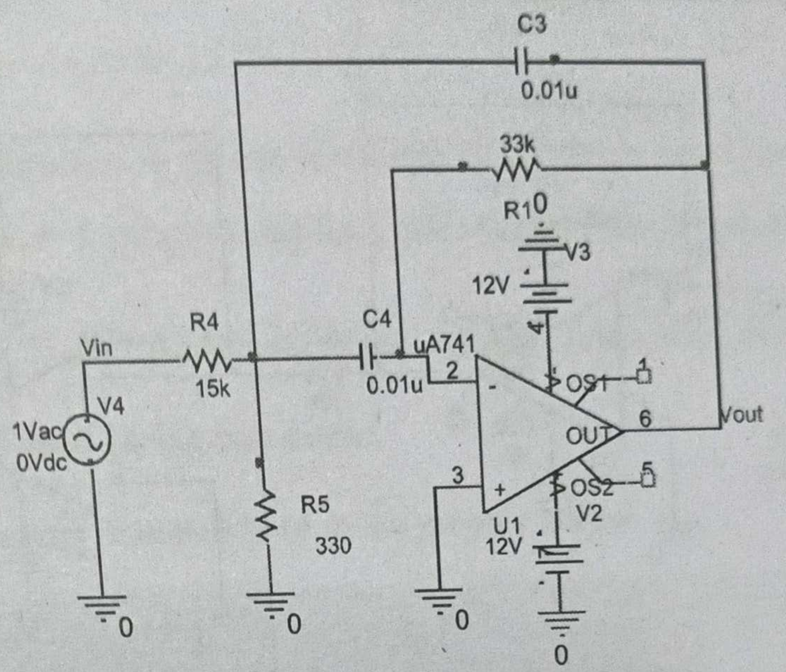
Circuit Diagram:



inverting schmitt trigger

Circuit Diagram:

Band Pass Filter



SIMULATION PROFILE

AC Sweep Time

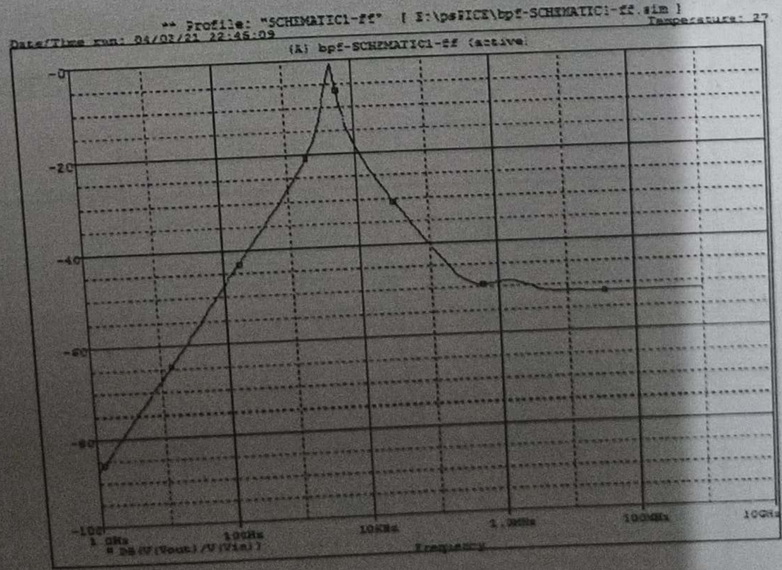
Logarithmic-Decade

Start Frequency =1

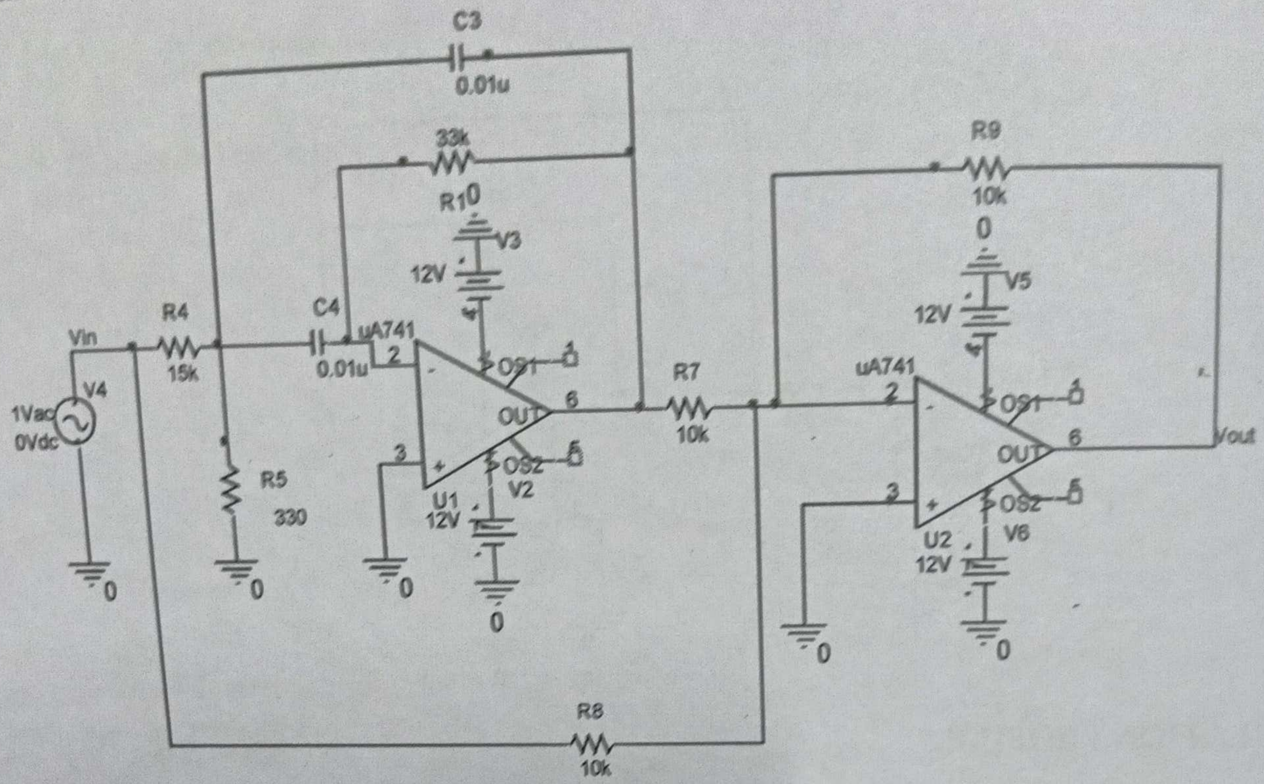
End Frequency=1G

Points/Decade=10

EXPECTED GRAPH



Band Reject Filter



SIMULATION PROFILE

AC Sweep Time

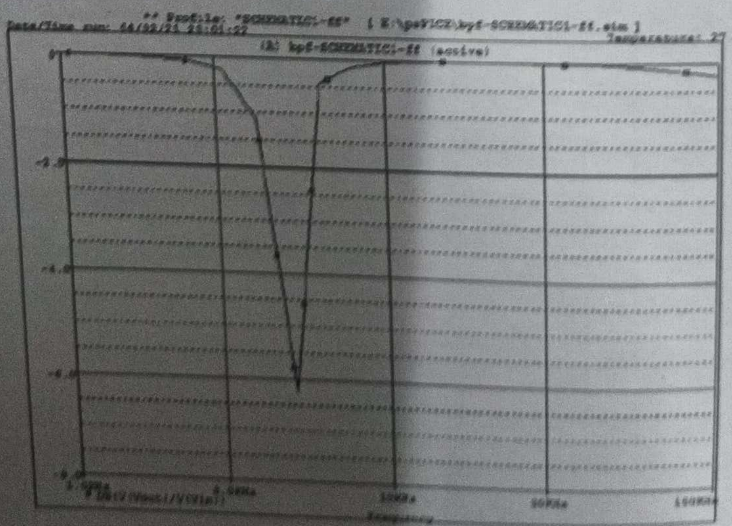
Logarithmic-Decade

Start Frequency =1

End Frequency=100k

Points/Decade=10

EXPECTED GRAPH



Experiment No:06 Date:

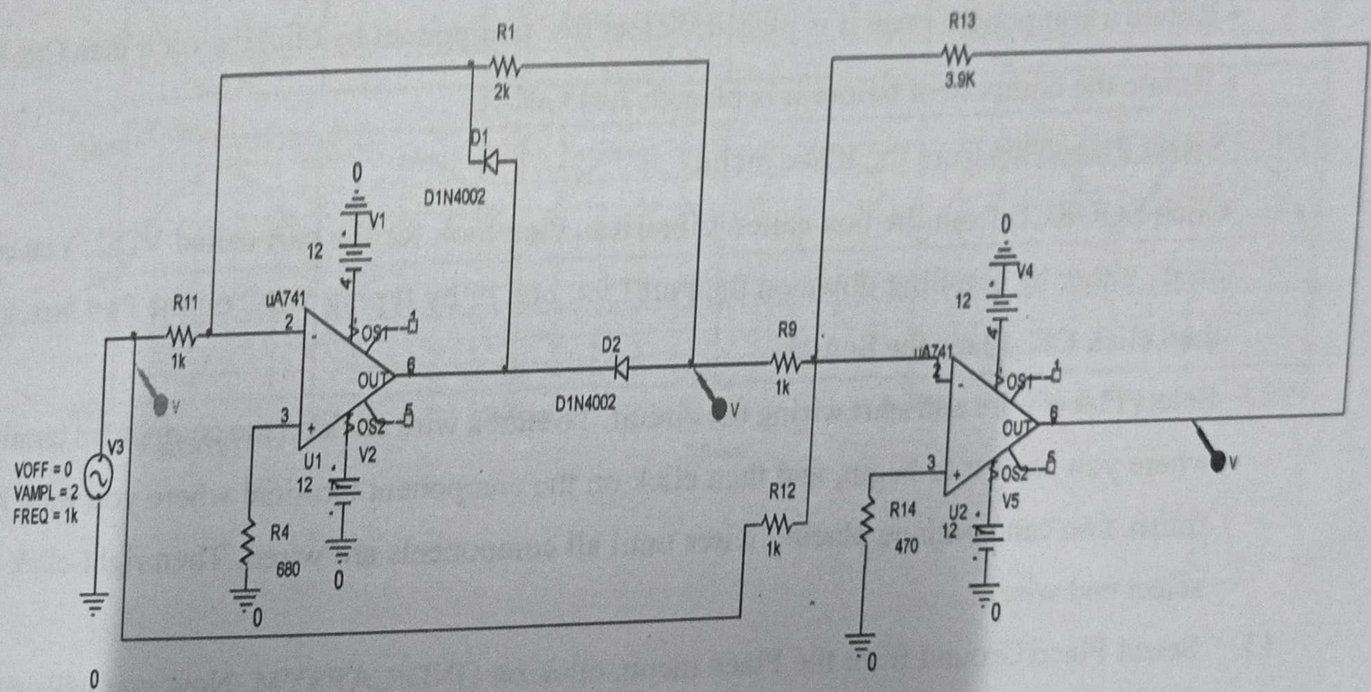
PRECISION RECTIFIERS

Aim: To simulate the performance of half wave and full-wave precision rectifier using op-amp

Theory:

The major limitation of ordinary diode is that it cannot rectify voltages below Vr(0.6v), the cut-in voltage of the diode. A circuit which is capable of rectifying input signals of the order of mv is called as precision rectifier.

Circuit Diagram:



Procedure:

Steps for Simulation of circuit using PSpice

1. Run the CAPTURE program.
2. Select File/New/Project from the File menu.
3. On the New Project window select Analog or Mixed A/D, and give a name to your project then click OK.
4. The Create PSpice Project window will pop up, select Create a blank project, and then click OK.